

All the **life phenomena** like nutrition, respiration, excretion, irritability, etc., are due to **interactions** between the various components of an organism. Properties of tissues are not present in all the constituent cells but arise as a result of interactions among the constituent cells. Similarly properties of cellular organelles are not present in the molecular constituents of the organelle but arise as a result of interactions among the molecular components comprising the organelle. Thus, interactions result in certain properties, at a higher level of organization. This phenomenon is true in the **hierarchy of organizational complexity** at all levels. **Biology is the story of life and evolution of living organisms on the Earth.** All living organisms- the present, past and future are linked to one another by sharing a common **genetic material**, but to varying degrees.

1.2 NATURE, SCOPE AND MEANING OF ZOOLOGY

Biology is a science devoted to the study of living organisms. Science progressed by breaking down complex subjects into their component parts and so today there are numerous branches of biology of which BOTANY, ZOOLOGY and MICROBIOLOGY are the principal, heterogeneous and divergent groups.

Zoology (zoon-animal; logos-knowledge / study) or '**Animal Science**' is a part of the biological science which deals with the study of various aspects of animals, starting from the sponges (Phylum : Porifera) to mammals (Class : Mammalia). The aim of zoology is to explain the **animal world** in terms of **scientific principles**.

Zoology is studied as a 'pure science' (knowledge gaining) and it has application in other branches such as euphenics, eugenics, biotechnology, bioenergetics, bioinformatics, etc. As applied science, it has tremendous scope in agriculture, aquaculture, animal husbandry, human health, diseases, veterinary science, apiculture, sericulture, pharmacology, animal breeding, etc.

1.3 BRANCHES OF ZOOLOGY

Lamarck (1809), a French biologist coined the term '**Biology**', which means the 'study of living organisms'. This diverse science which deals with all aspects of animal life has several sub-branches. A few of them are listed below.

Sub branches of Zoology

1. **Taxonomy** (Taxis - arrangement; nomos - rule, custom)

It is the theory and practice of identification, nomenclature and classification of organisms. The term 'Taxonomy' was coined by A.P. de Candolle.

2. **Morphology** (morphos - form; logos - study)

It deals with the study of form, size, shape, colour and structure of various organisms and their tissues, organs, organ- systems, etc. It includes the following

- i. **External morphology** It is the study of external characters of an organism.
- ii. **Internal morphology** This branch deals with the study of internal structure. It includes the following
 - a) **Anatomy** (ana - up; tome - cutting): It is the study of the internal arrangement of different organs or organ systems in an organism as observed with the naked eye.
 - b) **Histology** (histos - tissue; logos - study): It is the study of microscopic structure of different tissues. This branch is also referred to as '**Microanatomy**'.
3. **Cytology** (Kytos - cells; logos - study):

Cytology deals with the study of form and structure of cells and cell organelles. Cell biology is the branch of science that deals with the study of the cell as a structural and functional unit of living organisms.
4. **Physiology** (Physis - nature of functioning; logos - study):

It is the study of different body functions and processes.
5. **Embryology** (Embryon - embryo; logos - study) and **Developmental biology**

Embryology deals with the study of events that lead to fertilization, cleavages, early growth and differentiation of zygote into an embryo. It is also defined as the branch of biology dealing with the formation and development of embryos. **Developmental biology** is the study of embryonic development and the other developmental processes after birth.
6. **Evolution** (e - out; volva - roll):

It is the study of origin of life and continuous genetic adaptations of organisms to the environment. It also deals with the gradual changes that occur in the living organisms through geological time. Evolution means 'unfolding'. **Herbert Spencer** coined the term '**Organic Evolution**'.
7. **Palaeontology** (Paleo - ancient; on - being; logos - study)

Study of fossilized remains of organisms of the past geological ages is called palaeontology. This includes palaeobotany(fossils of plants) and palaeozoology (fossils of animals).
8. **Ecology** (Oikos - house; logos - study)

It is the study of living organisms in relation to the other living organisms (biotic factors) and abiotic environmental factors surrounding them. **Haeckel** coined the term '**Ecology**'.
9. **Genetics** (Gen - to grow into)

Genetics is the study of inheritance of characters from one generation to the next. It deals with **heredity** and **variations**. **Bateson** coined the term 'Genetics'.